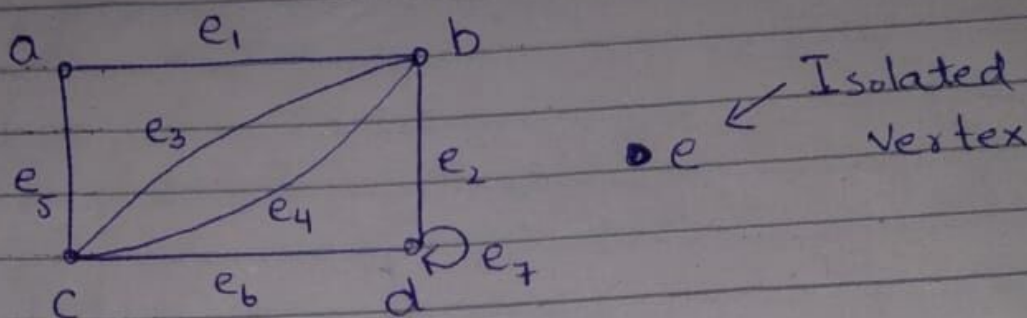


Graphs

$G(V, E)$

$V \Rightarrow$ Vertices / Nodes

$E \Rightarrow$ Edges



Here e_3 and e_4 parallel edges

e_7 loop edge

Adjacent vertices

$a \Rightarrow b, c$

$b \Rightarrow a, c, d$

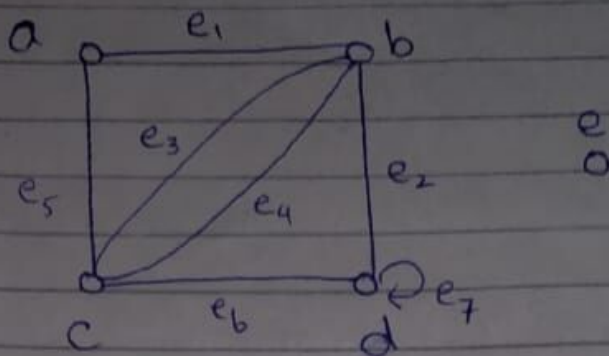
$d \Rightarrow b, c$

$V = \{a, b, c, d\}$

$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$

Degree of Vertices:-

(2)



$$\deg(a) = 2$$

$$\deg(e) = 0$$

$$\deg(b) = 4$$

$$\deg(c) = 4$$

$$\deg(d) = 2$$

Loop degree has "2"

→ e vertex isolated (یک ایسول)

Types of Graph :-

① Simple Graph

② Planer //

③ Regular //

④ Bipartite //

⑤ Complete //

⑥ Multi //

⑦ Directed Graph

③

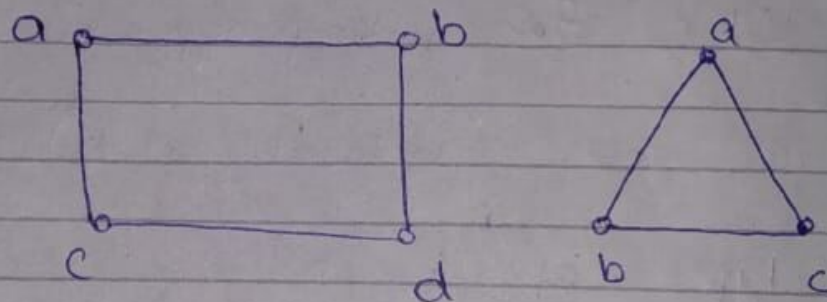
⑧ Undirected //

Now we discuss each graph with details.

1) Simple Graph

The graph which has no loops or parallel edges.

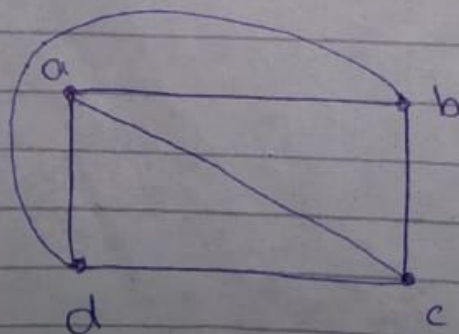
Examples:-



2) Planer Graph

An planer Graph, there is no cross edges.

Example :-



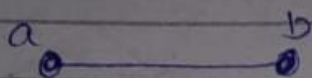
③ Regular Graph

The graph that have equal degree of each edge/node.

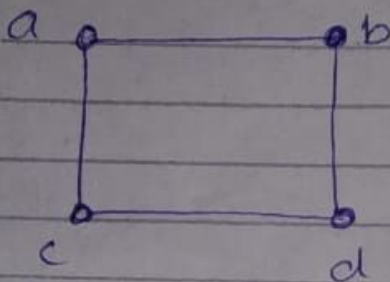
Examples:-



0- Regular

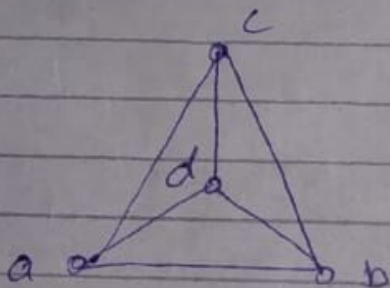


1- Regular



2- Regular

b/c each edge have same degree



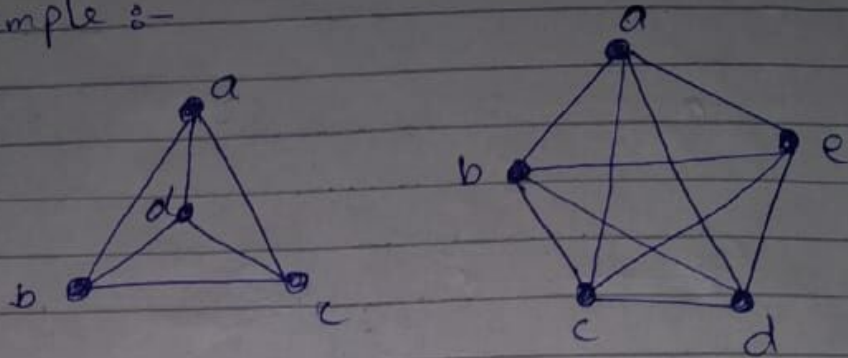
3- Regular

⑤

④ Complete Graph :-

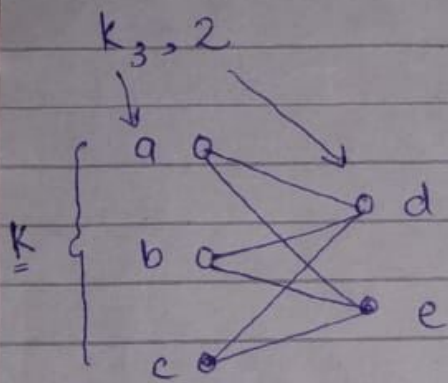
In this graph, each ^{Node} edge connects to all other edges/Nodes

Example :-

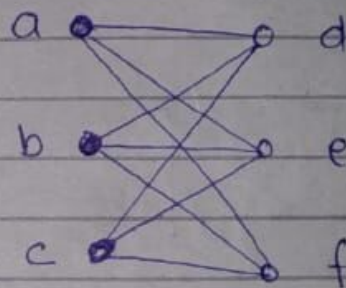


⑤ Bipartite Graph :-

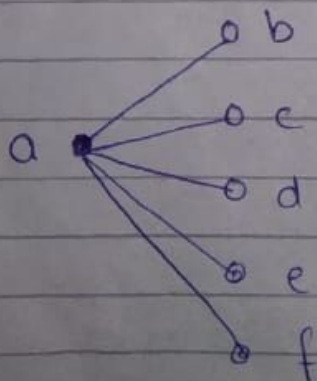
Examples :-



$K_{3,3}$



$K_{1,5}$

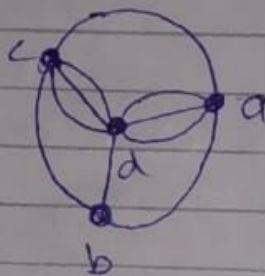
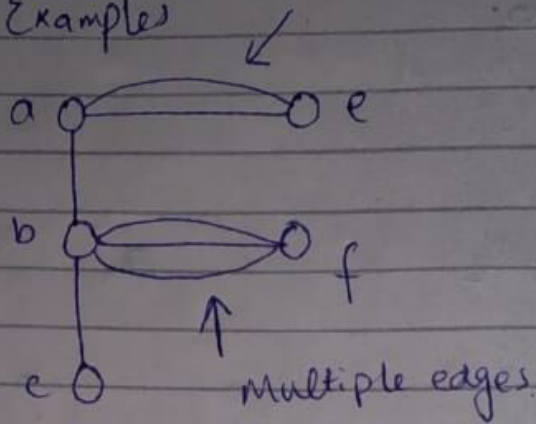


⑥

⑥ Multigraph:-

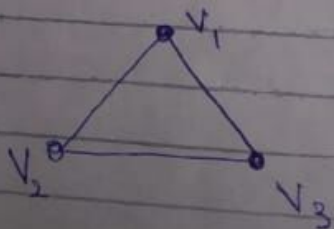
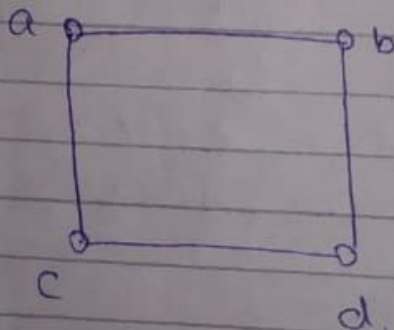
A graph is a multigraph which is permitted to have multiple edges, edges that have same end nodes.

Examples



⑦ Undirected Graph

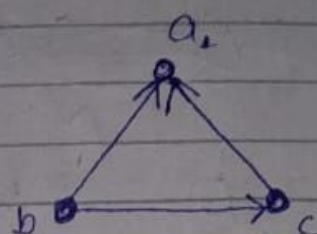
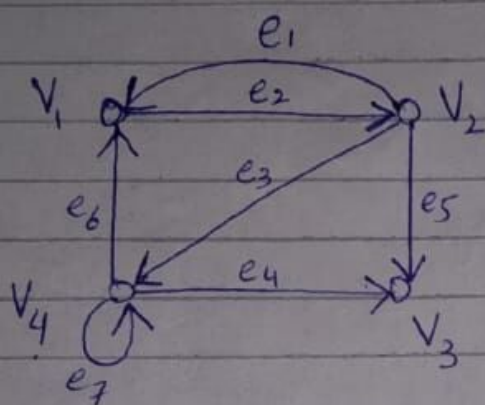
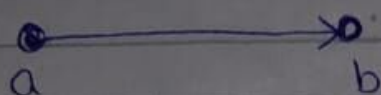
... have no direction of edges.



③ Directed Graph :-

In directed graph, there is a direction from start node to end node.

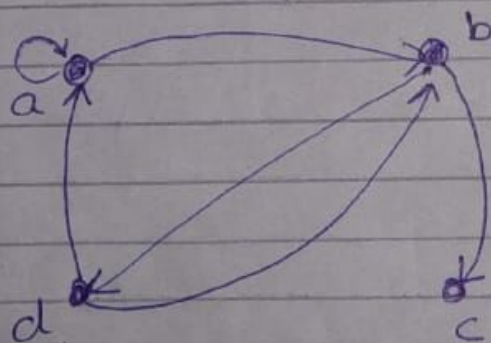
Examples:-



Degree In and Degree Out :-

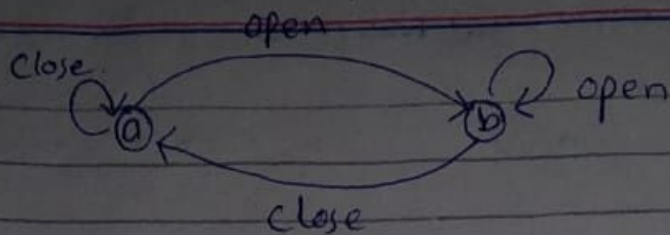
The degree in and degree out concept lies in directed graphs.

Examples:-



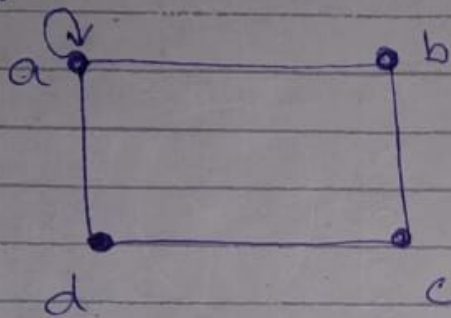
Node	In	Out
a	2	1
b	2	2
c	1	0
d	1	2

Note :- The degree of loop has one in and one out.



Vertex	In	Out
a	2	2
b	2	2

Matrix Representation of Graph :-
e.g 1:-



$$A = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Adjacency Matrix.

Inverse :-

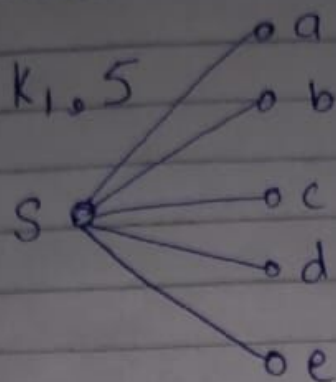
Adjacency list

$$A^{-1} = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

a	a, b, d
b	a, c
c	b, d
d	a, c

Matrix Representation of Graphs

Example 2:-



Adjacency Matrix:-

	s	a	b	c	d	e
s	0	1	1	1	1	1
a	1	0	0	0	0	0
b	1	0	0	0	0	0
c	1	0	0	0	0	0
d	1	0	0	0	0	0
e	1	0	0	0	0	0

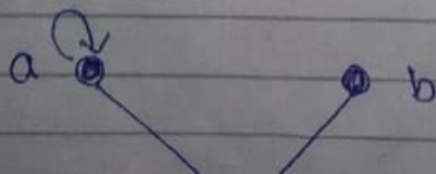
Conversion from Matrix to Graph:-

Examples:-

We have a matrix.

	a	b	c
a	1	0	1
b	0	0	1
c	1	1	1

We made three nodes/vertices.

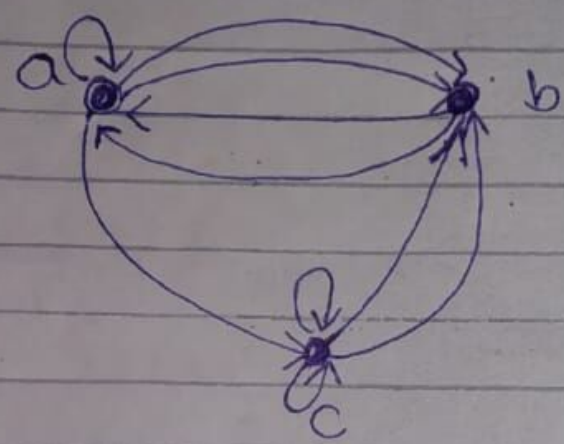


8:00 AM
12:00 PM
1:00 PM

(6)

Example 2:- Conversion

	a	b	c
a	1	2	1
b	2	0	0
c	0	2	2



Isomorphism Graph

lec #16

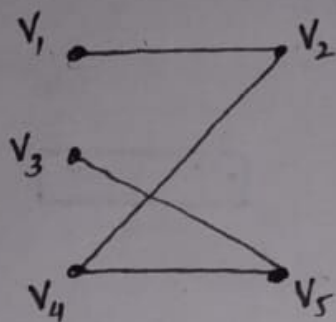
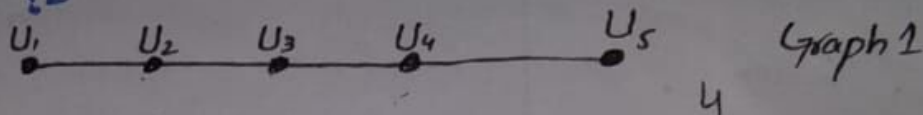
1

Iso $\Rightarrow$ Same

Two conditions

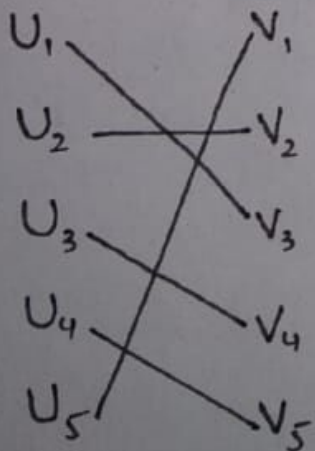
- ① No. of vertex are same.
- ② No. of edges of these two graphs are Same.

Examples :-



Check Conditions:-

- 1) No. of vertices of these two graphs are same i.e. = 5
- 2) No. of edges are same i.e. = 4



So these are
Isomorphism.